Fossil Fuels

Exam code: 0620 & 0971

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Updated on 1 September 2025

Common fossil fuels

- A fuel is a substance which when burned, releases heat energy
- This heat can be transferred into electricity, which we use in our daily lives
- Most common fossil fuels include **coal**, **natural gas** and **petroleum**
- **Methane,** CH_4, is the main constituent of natural gas and is a hydrocarbon
- **Hydrocarbons** are made from hydrogen and carbon atoms only

Petroleum & fractional distillation

Petroleum

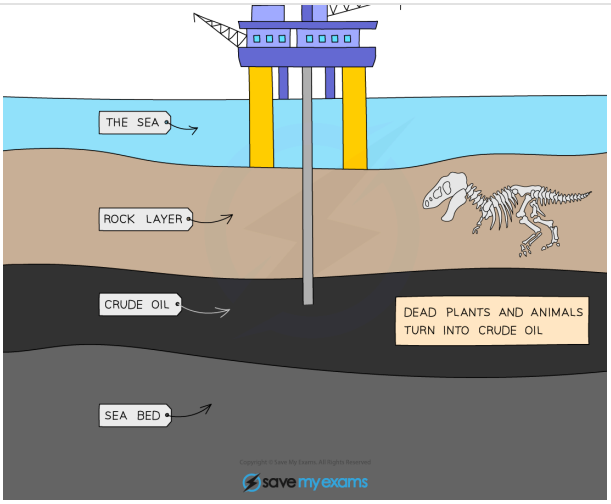
- Petroleum is also called **crude oil** and is a mixture of hydrocarbons which also contains natural gas
- It is a thick, sticky, black liquid that is found under porous rock (under the ground and under the sea)

The location of crude oil

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



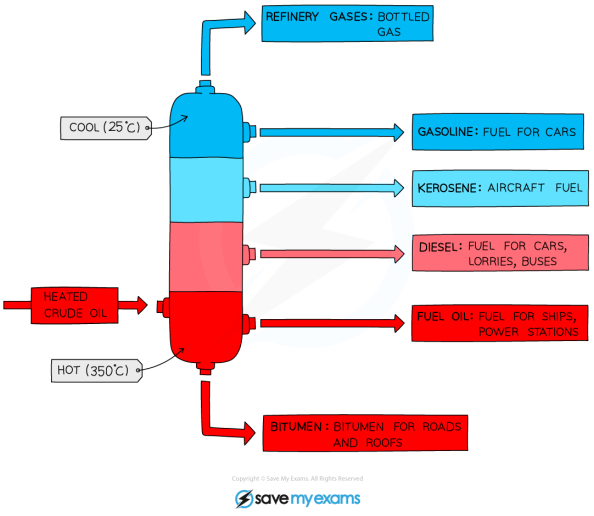
Crude oil is located under the sea

- Petroleum itself as a mixture isn't very useful but each component part of the mixture, called a **fraction**, is useful and each fraction has different applications

Fractional distillation

- Each fraction consists of groups of hydrocarbons of **similar** chain lengths
- The fractions in petroleum are separated from each other in a process called **fractional distillation**
- The molecules in each fraction have similar **properties** and **boiling points**, which depend on the number of carbon atoms in the chain
- The size of each molecule is directly related to how many carbon and hydrogen atoms the molecule contains
- Most fractions contain mainly **alkanes**, which are compounds of carbon and hydrogen with only **single** bonds between them

Diagram to show fractional distillation



Help me understand

- Fractional distillation is carried out in a **fractionating column** which has a temperature gradient
  - It is very hot at the bottom of the column and cooler at the top

- During the process of fractional distillation:
  - The crude oil is heated and **vapourises**
  - The vapours of hydrocarbons enter the column which has a **temperature gradient**
  - The vapours of hydrocarbons with high boiling points **condense** at the bottom of the column
  - The vapours of hydrocarbons with lower boiling points rise up the column and condense at the top

Properties of fractions

- **Viscosity**
  - This refers to the ease of flow of a liquid.
  - High viscosity liquids are thick and flow less easily.
  - As the number of carbon atoms increases, the attraction between the hydrocarbon molecules also increases which results in the liquid becoming more viscous with the increasing length of the hydrocarbon chain.
  - Going down the column, the viscosity of the fractions increases

- **Melting point/boiling point**
  - As the molecules get larger, the intermolecular attraction becomes greater.
  - More heat is needed to separate the molecules.
  - Going down the column, the boiling points of the fractions increases

- **Volatility**
  - Volatility refers to the tendency of a substance to vaporise.
  - As the size of the hydrocarbon increases, the attraction between the molecules increases
  - Going down the column, the volatility of the fractions therefore decreases

Uses of Fractions

- Refinery gas: heating and cooking
- Gasoline: fuel for cars (petrol)
- Naphtha: raw product for producing chemicals
- Kerosene: for making jet fuel (paraffin)
- Diesel: fuel for diesel engines (gas oil)
- Fuel oil: fuel for ships and for home heating
- Lubricating oil: for lubricants, polishes, waxes
- Bitumen: for surfacing roads

Table to show the trends in properties

Fraction	Number of carbon atoms	Boiling point range / °C	Viscosity	Volatility

Gasoline / petrol	4–12	40–100	down the fractions	down the fractions
Naptha	7–14	90–150		
Kerosene / paraffin	12–16	150–240		
Diesel / gas oil	14–18	220–300		
Fuel oil	19–25	250–320		
Lubricating oil	20–40	300–350		
Bitumen	More than 70	More than 350		

Examiner Tips and Tricks

You **must** be able to recall a use for each fraction as well describe the trends in properties.

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